Tau-Squared Estimation

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Introduction

The random-effects meta-analysis requires an estimate of τ^2, the between-study variance. Several estimators exist with different bias-variance trade-offs; the choice affects pooled estimates, CIs, and downstream inference. REML is the modern default; DerSimonian-Laird (DL) the historical default.

Prerequisites

Random-effects meta-analysis; method-of-moments vs maximum likelihood.

Theory

Common estimators: - **DerSimonian-Laird (DL)**: method-of-moments; fastest, historically default. - **REML (Restricted Maximum Likelihood)**: likelihood-based, generally less biased than DL; modern default. - **Paule-Mandel (PM)**: iterative method-of-moments; often similar to REML. - **Sidik-Jonkman (SJ)**: analytic, robust to model misspecification. - **Empirical Bayes (EB)**: iterative Bayes-style estimator.

DL systematically underestimates τ^2 under moderate heterogeneity; REML is close to unbiased.

Assumptions

True effects are drawn from a distribution (typically normal) with common variance τ^2; within-study variances are known.

R Implementation

```r
library(metafor)

set.seed(2026)
k <- 10
n_per <- sample(30:200, k, replace = TRUE)
true_theta <- rnorm(k, 0.3, 0.25)
yi <- true_theta + rnorm(k, 0, sqrt(4 / n_per))
vi <- 4 / n_per

methods <- c("DL", "REML", "PM", "SJ", "EB")
tau2_est <- sapply(methods, function(m) {
  rma(yi, vi, method = m)$tau2
})
round(tau2_est, 3)
```

Output & Results

τ^2 estimates under the five methods. DL typically gives the smallest value; REML, PM, SJ, EB are often close.

Interpretation

"τ^2 estimates ranged from 0.041 (DL) to 0.061 (SJ); we report REML ($\tau^2 = 0.055$, 95 % CI 0.012-0.142) as the primary estimator following modern meta-analysis recommendations."

Practical Tips

- Use REML by default; it is less biased than DL under most conditions.
- Hartung-Knapp adjustment (`test = "knha"` in metafor) improves CI coverage, especially with few studies.
- Compare multiple estimators as sensitivity; if DL and REML disagree substantially, heterogeneity is likely real.
- Report 95 % CI on τ^2 (profile likelihood or Q-profile methods) alongside the point estimate.
- For very few studies ($k < 5$), any τ^2 estimate is unreliable; use Bayesian methods with informative priors.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis